Second-Order Group Influence Functions

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Research Meeting

2026-02-27

Second-Order Taylor Expansion

$$f(\theta^*_{-\mathcal{S}}) - f(\theta^*) \approx \underbrace{-\varepsilon\, \boldsymbol{g}_f^\top \bar{\boldsymbol{v}}}_{I_1 \text{ (Koh \& Liang, 2017)}} + \frac{\varepsilon^2}{2}\Big[\underbrace{\bar{\boldsymbol{v}}^\top H_f\, \bar{\boldsymbol{v}}}_{T_1 \text{ (ours)}} + \underbrace{2\, \boldsymbol{u}^\top H_{\mathcal{S}}\, \bar{\boldsymbol{v}}}_{T_2 \text{ (Basu et al., 2020)}}\Big]$$

Notation: $\bar{\boldsymbol{v}} = H^{-1}\bar{\boldsymbol{g}}_{\mathcal{S}}$, $\bar{\boldsymbol{g}}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \nabla\ell(z_i, \theta^*)$, $\boldsymbol{u} = H^{-1}\boldsymbol{g}_f$, $\boldsymbol{g}_f = \nabla_\theta f(\theta^*)$

- T_2: couples test response $\boldsymbol{u}$ and influence direction $\bar{\boldsymbol{v}}$ via subset Hessian $H_{\mathcal{S}}$
- T_1: quadratic form of $\bar{\boldsymbol{v}}$ under test loss Hessian H_f

Goal: Predict Group Influence

Group influence of $\mathcal{S}$ on average test loss:

$$I_{\mathcal{S}} = f(\theta^*_{-\mathcal{S}}) - f(\theta^*) = \text{change in avg test loss when } \mathcal{S} \text{ is removed}$$

- **Goal**: build a score $\hat{I}_{\mathcal{S}}$ that ranks subsets by $I_{\mathcal{S}}$ accurately
- **Application**: identify training subsets most harmful/helpful to test performance

Evaluation Metric: LDS

Linear Datamodeling Score (LDS):

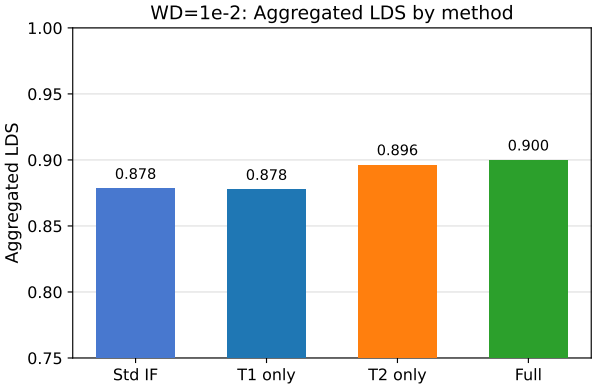
$$\text{LDS} = \rho\Big(\{\hat{I}_{\mathcal{S}}\}_{\mathcal{S}\in\mathcal{D}},\ \{I_{\mathcal{S}}\}_{\mathcal{S}\in\mathcal{D}}\Big) \quad \text{(Spearman over evaluation subsets)}$$

- $\hat{I}_{\mathcal{S}} = \frac{1}{|T|}\sum_t \hat{I}_{\mathcal{S}}^{(t)}$ (predicted group influence, averaged over test points)
- $I_{\mathcal{S}} =$ actual Δ (average test loss) when $\mathcal{S}$ is removed

Experiment Setup

- **Model**: Logistic Regression on MNIST (10-class), exact Hessian inversion (no iHVP approximation error)
- **Data**: 5,000 train / 2,500-sample subsets (50 random) / 100 test points

Main Results (50 Random Subsets)



Interpretation of Main Results

LDS values (50 random subsets):

Method	LDS
Std IF (I_1)	0.878
$I_1 + T_1$	0.878
$I_1 + T_2$	0.896
Full ($I_1 + T_1 + T_2$)	0.900

- T_1: marginal gain on its own in random subsets
- T_2: dominant 2nd-order correction (+1.8 pp over Std)

Negative Result: T_1 Suppressed in Random Subsets

- When sampling randomly, individual $\boldsymbol{v}_i$ point in diverse directions $\Rightarrow \bar{\boldsymbol{v}} = \frac{1}{|\mathcal{S}|}\sum_i \boldsymbol{v}_i$ is **likely small**
- $T_1 = \bar{\boldsymbol{v}}^\top H_f \bar{\boldsymbol{v}} \propto \|\bar{\boldsymbol{v}}\|^2 \Rightarrow$ strongly suppressed in expectation
- Next: intentionally sample subsets to amplify $\|\bar{\boldsymbol{v}}\|$ and examine how T_1 performs

Alternative: IF-Based Subset Selection

Individual first-order influence for selecting v_i:

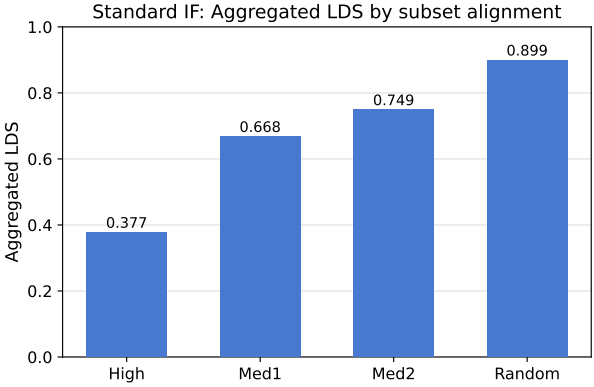
$$\mathrm{IF}_i^{(t)} = \boldsymbol{g}_t^\top \boldsymbol{v}_i \qquad \boldsymbol{v}_i = H^{-1}\nabla\ell(z_i, \theta^*)$$

- Select top-k training samples by $\mathrm{IF}_i^{(t)}$
- All selected v_i point toward $\boldsymbol{g}_t \Rightarrow \|\bar{\boldsymbol{v}}\|$ is amplified

High / Medium Alignment Settings

- **High**: sample from top-3,000 (60%) training pool $\Rightarrow$ most aligned v_i, $\|\bar{\boldsymbol{v}}\| \approx 2.45$
- **Med1 / Med2**: 70% / 80% pool $\Rightarrow$ moderate alignment, $\|\bar{\boldsymbol{v}}\| \approx 2.28$ / 2.19
- **Random**: uniform (existing EXP-10 setting), $\|\bar{\boldsymbol{v}}\| \approx 2.55$ (but direction random)

Standard IF Fails on IF-Selected Subsets



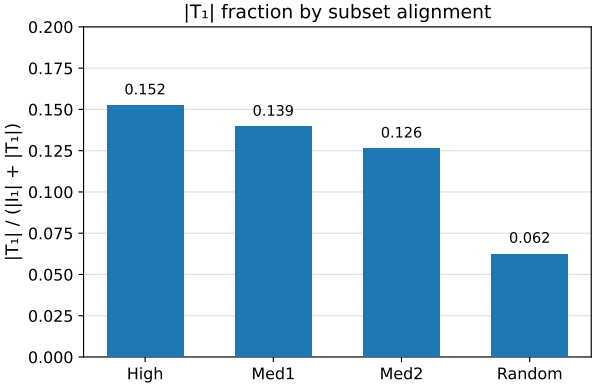
Our Correction is Consistent with Failure Mode

Std+T_1 improvement (LDS):

Setting	Std IF	Std+T_1	Δ
High	0.377	0.486	+10.8 pp
Med1	0.668	0.696	+2.9 pp
Med2	0.749	0.774	+2.6 pp
Random	0.899	0.902	+0.3 pp

- Largest gain *precisely where* Std IF fails most

T_1 Term Fraction vs. Setting

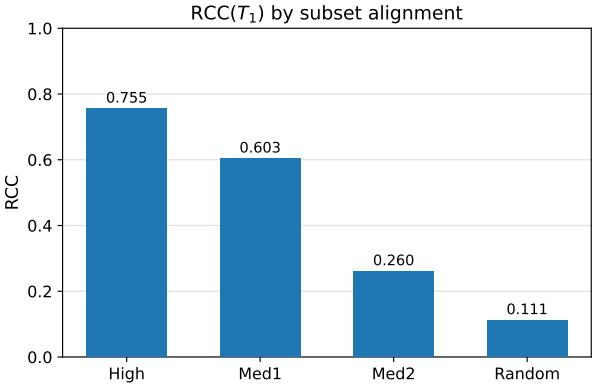


RCC: Residual Correlation Coefficient

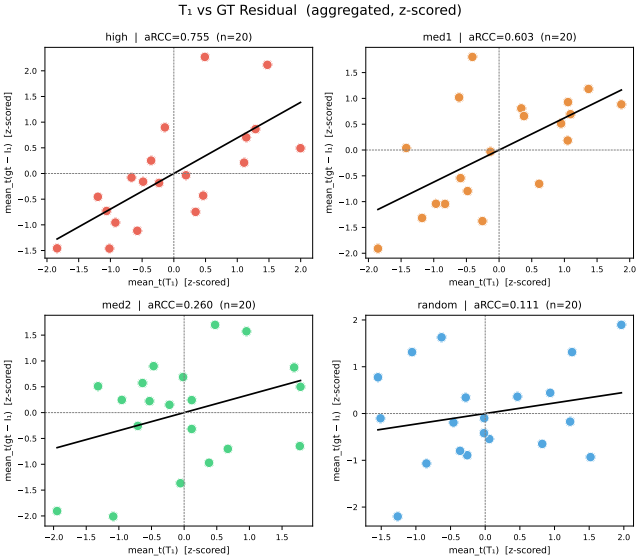
$$\mathrm{RCC}(X) = \rho\Big(\{X_{\mathcal{S}}\}_{\mathcal{S}\in\mathcal{D}},\ \{\mathrm{GT}_{\mathcal{S}} - I_{1,\mathcal{S}}\}_{\mathcal{S}\in\mathcal{D}}\Big) \quad \text{(Spearman over subsets)}$$

- Measures how well term X explains the *first-order residual* at the **group level**
- High $\mathrm{RCC}(T_1) \Rightarrow T_1$ provides signal beyond I_1

RCC Results


RCC(T_1) by subset alignment
RCC
1.0
0.8
0.6
0.4
0.2
0.0
0.755
0.603
0.260
0.111
High
Med1
Med2
Random

T_1 Correlates with First-Order Residual



T_1 Contains Genuinely Different Information

When $H_S \approx H$ (logistic regression):

$$T_2 = 2\,\boldsymbol{u}^\top H_S\,\bar{\boldsymbol{v}} \approx 2\,\boldsymbol{g}_f^\top\,\bar{\boldsymbol{v}} = -\tfrac{2}{\varepsilon}\,I_1$$

- T_2 and I_1 carry the *same* ranking information
- $T_1 = \bar{\boldsymbol{v}}^\top H_f\,\bar{\boldsymbol{v}}$ is **genuinely new signal**, orthogonal to I_1

Theoretical Direction 1: T_1 as Quality$\times$Diversity Penalty

Proposition (Helpfulness-Diversity Decomposition): Decompose $v_i = \alpha_i \hat{g}_f + v_i^{\perp}$ ($\alpha_i = \hat{g}_f^{\top} v_i$: helpfulness, $v_i^{\perp} \perp g_f$, $\bar{\alpha} = \frac{1}{|S|} \sum_i \alpha_i$).

$$I_1 = -\|g_f\| \, \bar{\alpha} \qquad \text{(depends only on mean helpfulness)}$$

$$T_1 = \bar{\alpha}^2 \lambda_{g_f} + 2\bar{\alpha} \, \hat{g}_f^{\top} H_f \bar{v}^{\perp} + (\bar{v}^{\perp})^{\top} H_f \bar{v}^{\perp} \geq 0 \qquad \text{(depends also on } \bar{v}^{\perp}\text{)}$$

- I_1: determined by $\bar{\alpha}$ alone — blind to orthogonal component
- T_1 penalty amplified by $\|\bar{v}^{\perp}\|$: diverse subset $\Rightarrow$ $\bar{v}^{\perp} \approx 0$ $\Rightarrow$ penalty minimized while preserving helpfulness

Interpretation: Quality Alone Is Not Enough

- Selecting top-k by I_1 (quality) aligns all v_i toward $g_f \Rightarrow \bar{v}^{\perp}$ can be large $\Rightarrow T_1$ penalty grows
- **Implication**: best subset needs both high quality ($\bar{\alpha}$ large) *and* diversity ($\bar{v}^{\perp}$ small)
- **Validation**: subset selection experiment — compare quality-only vs. quality+diversity selection on actual test loss improvement

Theoretical Direction 2: T_1 Activated by Regularization

Proposition (Training Confidence $\rightarrow$ Term 1 Vanishing):

$$|I_1| = O(\delta_{\mathcal{S}}), \quad |T_1| = O(\delta_{\mathcal{S}}^2) \quad \text{where } \delta_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} |p_i - y_i|$$

- Without weight decay: $\delta_{\mathcal{S}} \rightarrow 0 \Rightarrow T_1$ vanishes *faster* than I_1
- With weight decay $\lambda_{\text{wd}} > 0$: $\delta_{\mathcal{S}} \geq c'(\lambda_{\text{wd}}) > 0 \Rightarrow$ vanishing blocked, T_1 remains active

Interpretation: When Does T_1 Matter?

- Confident training ($\delta_{\mathcal{S}} \approx 0$): $|T_1|/|I_1| \approx 0$ — first-order is sufficient
- Regularized training ($\delta_{\mathcal{S}}$ bounded away from 0): $|T_1|/|I_1|$ grows — second-order correction matters
- **Validation**: sweep weight decay, measure $|T_1|/(|I_1| + |T_1|)$ ratio — expect monotone increase with $\delta_{\mathcal{S}}$